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Short- and Long-Term Effects of Repetitive Mother–Infant Separations on Social Development in Rhesus Monkeys

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Rhesus monkey mother–infant dyads were each subjected to sixteen, 4-day physical separations between the infants' 3rd and 9th months of life. The infants displayed protest behavior following each separation but only minimal signs of despair. Their protest diminished somewhat over repeated separations. The mothers' separation reactions were considerably milder (and changed little) over repeated separations. The separations appeared to retard the development of normal mother–infant relationships: Relative to nonseparated control dyads, separated infants displayed excessive levels of infantile behaviors, although their mothers did not differ from control mothers in levels of any behavior. Near the end of their 1st year, all infants were permanently separated from their mothers and housed as peer groups. Over the next 30 weeks during peer housing, few behavioral differences emerged between previously separated and control subjects. However, when exposed to their mothers during preference tests, previously separated monkeys seemed to avoid their mothers in sharp contrast to the mother-seeking activity displayed by control infants.

The study of mother–infant separation has long attracted a wide range of researchers interested in the process of social development in humans as well as nonhuman primates. Mother–infant separation studies have been carried out for a variety of reasons, including furthering the understanding of the dynamics underlying the attachment process (e.g., Ainsworth, 1978; Bowlby, 1969, 1973; Hinde, 1974) and the role that mother–infant separation may play as a possible cause of both short-term and long-term psychopathology (e.g., Bowlby, 1960, 1973; Harlow & Suomi, 1974; Kaufman & Rosenblum, 1967; Spitz, 1946). Recently, very real practical concerns have also contributed to interest in the consequences of mother–infant separation. These include the possible effects of frequent brief maternal separations, such as occur in day-care programs, as well as the effects of longer term or permanent maternal separations, such as occur in "shared" parenthood and paternal custody arrangements in divorce settlements.

There is now considerable evidence that when most human and nonhuman primate infants are forcibly separated from their mothers, they quickly become agitated and distressed, a reaction first termed *protest* by Robertson and Bowlby (1952). Most researchers studying the phenomenon have considered protest to be a universal initial reaction to maternal separation for primate infants, probably the product of natural selection (Kaufman & Rosenblum, 1967; Rosenblum & Plimpton, 1980). In many primate species, a second stage of reaction to maternal separation has been reported for some infants. This second stage, termed *despair* by Bowlby (1960), appears to be highly depressive in nature, in that behavioral and physiological concomitants of infant despair clearly resemble certain clinical symptoms of severe human depression (Mineka, 1982; Mineka & Suomi, 1978; Reite, Short, Seiler, & Pauley, 1981; Spitz, 1946). At the present

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time the basis for the large degree of between- and within-species differences in this separation-induced despair reaction is not fully understood, although factors associated with an infant's preseparation attachment relationship with its mother and the nature of its separation environment are clearly implicated (Mineka & Suomi, 1978; Suomi, 1976).

One issue in the mother–infant separation literature that has received relatively little attention but that is of concern in the present article is the effect of repeated separations. Of the few studies that have examined the effects of more than one mother–infant separation, similar responses to each of the repeated separations have been reported (e.g., Chappell & Meier, 1975; Coe & Levine, 1981; Hinde & Spencer-Booth, 1971). However, in each of these studies, the number of repeated separations was few (i.e., <4), and the time between separations was considerable (i.e., >1 week).[1] By contrast, several studies involving repetitive *peer* separations (in which subjects were separated from age-mates rather than from mothers) have suggested that significant changes can occur when there are many separations more closely spaced in time. For example, Suomi, Harlow, and Domek (1970) reported that peer-reared rhesus monkeys subjected to a total of twenty, 4-day separations between 3 and 9 months of age displayed some decline in the intensity of protest over the repeated separations, although the despair reaction seemed quite stable. More recently, Mineka, Suomi, and DeLizio (1981) also reported that substantial changes in response to separation occurred for 2- and 4-year-old rhesus monkeys subjected to 8 weekly 4-day peer separations. Specifically, protest reactions gradually disappeared over the repeated separations, whereas despair responses increased in intensity over the same period for these juvenile and adolescent subjects. These studies suggest that reactions to many closely spaced peer separations may change over time and that the nature of the changes may be different for monkeys of different ages. Unfortunately, at present no comparable evidence exists as to whether any such changes would result from repetitive separations of several days' duration at short intervals involving mothers rather than peers.

Determination of possible changes in the response to repetitive mother–infant separations is of interest for theoretical as well as practical reasons, since different theoretical perspectives on the nature of mother–infant attachment and separation make different predictions regarding such changes. An integration of reactance and learned helplessness theories, for example, would predict a decline in the protest response over repeated separations concomitant with an increase in the despair response (Mineka & Suomi, 1978; Seligman, 1975; Wortman & Brehm, 1975). Opponent-process theory also predicts an increase in the intensity of the despair response due to a growing beta-process with repeated separations (cf. Hoffman & Solomon, 1974; Mineka et al., 1981; Solomon, 1980) and can account for (if not predict) a decline in the protest response. By contrast, social learning approaches to attachment and separation (e.g., Cairns, 1977) would predict that infants should show habituation of any response that occurs to repeated separations (unless the protest response is reinforced by eliciting reunion behavior from parents, in which case protest would rise, e.g., Gewirtz & Boyd, 1977). Finally, an ethological attachment-theory perspective (e.g., Ainsworth, 1978; Bowlby, 1969, 1973) does not seem to make explicit predictions regarding changes in the nature of the response to repetitive separations, since the response is thought to represent a prepotent, instinctive reaction.

A second issue of concern for the present article is the possible long-term consequences of one or more separations—an issue that has received somewhat more attention in the mother–infant separation literature. For example, Spencer-Booth and Hinde (1971)

[1] Two very recent studies reported by Coe, Wiener, and Levine (in press) did examine larger numbers of repeated mother–infant separations in squirrel monkeys; significant changes in behavioral reactions were observed over repeated separations. The first study involving six 1-hour separations conducted at 10-day intervals found that the frequency of distress vocalizations decreased over repeated separations. By contrast, the second study, involving twenty 30-minute separations at 2-day intervals, reported significant increases in distress vocalizations and activity over repeated separations. At present the actual basis for these seemingly opposite changes in response over many repeated brief separations in squirrel monkeys is not clear.

found that some rhesus infants had residual effects of one or two separations from mother 2 years later (see also Hinde, Leighton-Shapiro, & McGinnis, 1979). In a related vein, Suomi et al. (1970) found that repetitive peer separations in infant rhesus monkeys resulted in severe retardation of normal development, whereas Mineka et al. (1981) found that repetitive peer separations in juvenile and adolescent peer-reared monkeys resulted in long-term depressive consequences. Again, such results are of interest theoretically. An attachment-theory perspective explicitly predicts, for example, that mother-directed behavior by infants should intensify over repeated separations as the infants become "anxiously attached," and the long-term consequence should be increased infant dependence on the mother (Bowlby, 1973). Opponent-process theory, on the other hand, predicts a dampening of attachment behaviors both over repeated reunions and as a long-term consequence of the separations (Mineka et al., 1981). Finally, a social-learning perspective would also predict increased infant independence from the mother or a dampening of attachment as a consequence of repeated separations (Cairns, 1977).

Given the widespread incidence of repeated human mother–infant separations that routinely take place in our society today, it would be useful to have a better understanding both of possible changes in response to repeated separations and of possible long-term consequences. Unfortunately, adequately controlled prospective experimentation involving deliberate repeated separation of human mother–infant dyads presents very real ethical and practical problems. Because responses to mother–infant separation in rhesus monkey infants have many apparent parallels to the responses of human infants (Bowlby, 1973; Hinde, 1974; Mineka & Suomi, 1978), it seemed worthwhile to pursue this issue with rhesus monkey mother–infant pairs.

In the present study, we examined the consequences for infant rhesus monkeys of two cycles of 8 weekly 4-day separations from their mothers, thus paralleling the repeated separation paradigm of Suomi et al. (1970) for infant peer-reared monkeys and Mineka et al. (1981) for juvenile and adolescent peer-reared monkeys. This paradigm was selected because those studies found significant changes in response to repetitive separations, as well as clear-cut, long-term consequences of the separations. In the present study, both infants' and mothers' responses to the repeated separation–reunion cycles were closely monitored. A control group of mother–infant pairs was observed over a comparable period of development, but not subjected to any separations. After the separations, long-term consequences were examined in several ways. First, both the previously separated and the control groups of mothers and infants were observed for an additional 6-week period in order to assess any residual effects of the repeated separations in these dyads. Then, when they were approximately 10½ months old (an age by which most rhesus monkey infants growing up in the wild have become largely independent of their mothers, e.g., Hinde, 1974), both previously separated and control infants were permanently separated from their mothers and assembled into peer groups. Subjects were maintained and observed in their respective peer groups for the next 30 weeks. Finally, at approximately 1½ years of age, all subjects were administered social preference tests to assess their relative preferences for their mothers when given the chance to choose among several familiar and unfamiliar conspecifics.

Method

Subjects

The subjects were eight rhesus monkeys and their mothers, assigned to two different groups of four mother–infant pairs according to order of birth. Four male infants with an age range of 40 days composed the separation (S) group. Two male and two female infants with an age range of 70 days made up the nonseparated control (C) group.

All eight mothers were individually housed in 1 × 1 × 1 m wire-mesh cages throughout their pregnancies; each had previously given birth to at least one other infant and had demonstrated species-normative maternal care. All births took place without complication in the mothers' respective home cages, and all mothers nursed their offspring without any difficulty, gradually weaning them beginning in the 2nd month of life. All mother–infant pairs remained undisturbed in their home cages (except for routine cage cleaning during which each pair was housed together in a transport cage) until the S subjects' first separation. Throughout this period and the rest of the study, all subjects were living under a daily 12 hour:12 hour light–dark cycle.

Procedure

Separation–reunion cycles. When the mean age of the S subjects was 14 weeks, the first series of eight separations and reunions was begun. During each separation week, each S infant was physically separated from its mother by the animal caretaking staff at 9:00 A.M. on Monday morning and taken to another room in the laboratory where it was housed individually in an .8 × .6 × .8 m wire-mesh cage for a 96-hour period. Other monkeys were housed in the same room during the separation. During each separation, mothers remained in their home cages. At 9:00 A.M. on Friday, each subject was returned to its home cage and reunited with its mother. This procedure was continued for a total of 8 weeks, with 4 days of separation and 3 days of reunion each week (Sep Series I).

After the eighth separation and reunion, all S infants and their mothers remained together in their home cages for a total of 6 weeks (Postsep I). At the end of this period, when the mean age of the S subjects was 28 weeks, a second series of 8 weekly 4-day-separation, 3-day-reunion cycles was begun (Sep Series II). The same basic procedure and schedule of separations and reunions was employed as in the first separation–reunion series. Following the last separation and reunion, S monkeys and their mothers continued to be housed together in their home cages until the infants' mean age was 42 weeks (Postsep II). With no separations for their first 42 weeks of life, control infants and mothers were housed together in their home cages in the same room in which the home cages of S mothers and infants were located.

Final separation and peer group housing. When the mean age of both S and C monkeys was 43 weeks, each subject was physically separated from its mother by caretaking staff. The four S monkeys were then placed in a 2.0 × 1.5 × 1.0 m group cage, whereas the four C monkeys were all placed in a second group cage of like dimensions. Both groups were maintained in their respective cages for the next 30 weeks, at which time the study was formally ended.

Social preference tests. Twenty-five weeks after peer group formation, each subject was briefly removed from its respective quad cage and given a 15-minute social preference test in the Sackett Self-Selection Circus (Sackett, 1970). Numerous studies conducted in this apparatus have previously assessed relative preferences of young macaques for familiar versus unfamiliar and kin versus unrelated conspecifics (cf. Sackett, 1970). In the first test, the subject's biological mother and three unfamiliar adult females (the biological mothers of the other three members of its peer group) served as stimulus animals. After a 5-minute *adaptation* period during which the subject could see and hear all the stimulus animals, the Plexiglas panels that confined it to the "start" part of the apparatus were raised, giving it access to the compartments adjacent to each of the four stimulus monkeys for a 10-minute *choice* period. A thorough description of this standard circus test procedure can be found in Sackett (1970).

One week after the first preference test, each subject was given a second 15-minute circus test. This time the subject's biological mother and each of the three peers in its social group served as stimulus animals; otherwise, the test procedure was exactly as described previously. In this second round of preference tests, each subject served as a stimulus animal three times in addition to its own 15-minute test, but never on the same day as its test. Order of testing within each peer group was randomly determined. All circus tests were carried out between 12:00 P.M. and 3:00 P.M. on each day of testing.

Data Collection

Observational data. Four testers with extensive previous experience in observational scoring of rhesus monkey behavior recorded the activities of each subject throughout the experiment. Additionally, observational data were collected on each subject's biological mother until the final maternal separation and peer-group formation.

For S group subjects and their mothers, observation sessions began 2 weeks prior to the first separation. Each observational session was 5 minutes long; during that time the frequency and duration of behaviors encompassed by each of 14 predefined behavioral categories were recorded by a single observer. Operational definitions of each category are presented in Table 1; together the categories formed an exhaustive set (cf. Sackett, 1978). The four observers were given reliability tests monthly; for no category did Pearson correlations fall below +.90 on any of these reliability tests.

Separation group infants and their mothers were each scored between 9:00 A.M. and 10:30 A.M. five times per week for the 2 weeks prior to their first separation. During each 4-day separation, subjects and their mothers were scored twice daily (Mondays through Thursdays) between 9:00 A.M. and 10:30 A.M. and between 11:30 A.M. and 1:00 P.M. During reunion periods (Fridays through Monday mornings), each S subject and its mother were scored three times, twice on Friday (between 9:00 A.M. and 11:30 A.M. and between 10:30 A.M. and 1:00 P.M.) and once on Monday prior to the next separation (between 8:00 A.M. and 9:00 A.M.). During Postsep I (after the first eight separation and reunion cycles) and Postsep II (after the second eight separation and reunion cycles), S subjects and their mothers were scored according to the same schedule as during the 2-week preseparation period (five times/week).

Observations of C-group infants and their mothers did not begin until the infants were on average 18 weeks old.[2] From 18 to 26 weeks, each C subject and its mother were scored twice weekly between 9:00 A.M. and 10:30 A.M. From weeks 27 to 42, each C infant and its mother were scored 4 days/week during the same hours.

Following permanent separation from their mothers (at 43 weeks), all S and C monkeys were each observed four times/week in their respective peer groups between

[2] Observational scoring of C infants and their mothers did not begin at 12 weeks, as did those of S infants and mothers, because at that time C infants and mothers were serving as control subjects in another study and were being scored using a different scoring system. This other study was completely noninvasive in nature and did not compromise these monkeys for the present study (see Footnote 3). It ended when C infants were on average 16 weeks of age.

Table 1
Operational Definitions of Behavioral Scoring Categories

Category	Definition
Mother–infant ventral	Mutual ventral contact and/or nipple contact (includes hovering over or cradling).
Mother–infant reject	Any successful or unsuccessful break of mother–infant ventral, or rejection of an initiated mother–infant ventral.
Ventral cling	Contact of own ventral body surface with another subject (excluding mother–infant ventral).
Self–clasp	Clasping of any part of own body with hand(s) and/or foot (feet).
Self–mouth	Oral contact with own body.
Environmental exploration	Tactual and/or oral manipulation by subject of inanimate objects.
Passive	Absence of all social, exploratory, and locomotor activity. May be scored with self-categories and vocalizations.
Stereotypy	Patterned movements maintained in a rhythmic and repetitive fashion for at least three cycles (scored independently and noninclusive of locomotion).
Locomotion	Any self-induced change in location of self.
Vocalization	Any sound emitted by subject.
Huddle	A maintained, self-enclosed, fetal-like position.
Play and sex	Any sexual posturing directed toward another subject and/or any type of play activity including mutual or nonmutual rough and tumble play, with or without locomotion and/or chase and play face. Scored independently of locomotion and/or social contact.
Aggression	Vigorous and/or prolonged biting, hair pulling, clasping, accompanied by one or more of threat, barking, piloerection, or strutting.
Social contact	Contact and/or proximity (within 5 cm) with another subject (exclusive of mother–infant ventral, ventral contact, aggression, and play and sex).

9:00 A.M. and 12:00 P.M. for a 16-week period (Weeks 43–58). These observations were reduced to twice weekly (between 9:00 A.M. and 12:00 P.M. for the final 14 weeks of the study (Weeks 59–72).

Preference test data. Data from the circus preference tests were collected according to the standard procedure described by Sackett (1970). A trained observer watched the tests over closed-circuit TV and recorded the total frequency and duration of subject entries into each stimulus animal compartment. Data were collected only during the 10-minute choice period of each 15-minute test. Two different observers each scored approximately half the tests. No reliability tests were run between the observers; however, in previous reliability tests in the circus, these same two observers consistently obtained Pearson correlations greater than .99 for choice compartment measures.

Data Analysis

The data from this experiment were analyzed as follows: (a) Behavioral measures collected on S group infants and their mothers during preseparation baseline were compared with a normative data base for mother–infant interactions among like-reared mother–infant pairs in the Primate Laboratory in order to ascertain whether the S group infants and their mothers were "normal" prior to the separations. (b) Next, data collected from S group infants and their mothers during the two series of eight 4-day separations were subjected to category-by-category three-way repeated measures analyses of variance (ANOVAs) with separation series (first eight separations vs. last eight separations), weeks within series (1–8), and days within weeks (1–4) as the repeated measures, in order to detect possible changes in behavioral levels both within and between repeated separations. Parallel three-way repeated measures ANOVAs were run on reunion data from S group infants and their mothers; the data were collected during the 3-day reunion periods that followed each separation. (c) Observational data collected during the two postseparation periods (Postsep I and II) from both S and C group infants were subjected to category-by-category three-way repeated measures ANOVAs, with subject group (S vs. C) as an independent factor and with postseparation period (I vs. II) and weeks within period (1–6) as repeated measures in order to compare developmental changes in behavioral levels beween separated and nonseparated infants during these periods. Parallel ANOVAs were run on measures of S and C mothers, which were collected during the same postseparation periods. (d) Observational data collected on S and C infants during the first 30 weeks of peer-group living were subjected to category-by-category two-way repeated measures ANOVAs, with subject group (S vs. C) as an independent factor and four time periods as repeated measures in order to assess possible group differences and/or changes over time following permanent separation from mothers and peer-group formation. (e) Finally, the results of the two circus preference tests for each S and C subject were analyzed by repeated measures ANOVAs, with subject group (S vs. C) as an independent variable and type of stimulus animal (mother vs. other adult females; mother vs. peers) as a repeated measure. In these and all other ANOVAs, whenever a significant ($p < .05$) main effect or interaction was detected, Duncan tests (Duncan, 1955) were subsequently applied to the appropriate set of means.

Results

The results of this study revealed that (a) prior to the repetitive separations, S group infants and their mothers displayed levels of behavior well within the normative range shown by similarly housed mother–infant pairs in the Wisconsin rhesus monkey colony. (b) Infants displayed protest reactions, with evidence of mild subsequent despair, to each of the repetitive 4-day separations. However, protest appeared to decrease somewhat over repeated separations. Mothers displayed much milder reactions to the repeated separations, with little evidence of major change over repeated separations. (c) The 3-day reunion periods following each separation were characterized by very high levels of mutual ventral contact between mothers and infants, with little change over repeated separations. (d) S subjects displayed more infantile patterns of behavior after each series of repetitive separations than did nonseparated C subjects; these differences were especially pronounced during the second postseparation period. In contrast, S and C mothers displayed similar levels of behavior during both Postseps I and II. (e) There were relatively few major differences between S and C monkeys, following peer group formation. If anything, C monkeys displayed slightly more intense short-term reactions but somewhat better long-term adjustment to peer group living. (f) In contrast, preference tests yielded substantial differences between S and C monkeys: S monkeys appeared to avoid their mothers relative to preferences displayed by C monkeys. Details of these results follow.

Preseparation Behaviors

Behavioral category scores for each S subject and its mother, obtained during the 2-week period immediately prior to the first separation (Weeks 12–13 of life for the infants), were summed over the 2-week period, and for every category, mean scores per observational session were calculated for each subject and its mother. Because data collection was not begun for C infants and mothers until 18 weeks (see Footnote 2), between-group comparisons of preseparation behaviors were not possible. However, there existed a "data bank" of observations made the previous year on 15 other infants (9 males and 6 females) and their mothers who had been housed in the same room and observed according to the same schedule and scoring system as the S infants and their mothers. Mean category scores per session during Weeks 12–13 were extracted from the data bank for these 15 infants and their mothers, and these means were compared with those of the S group infants and their mothers by uncorrelated t tests. No significant differences between S group infants and data-bank infants were disclosed for any category; the same was true for comparisons of mothers' category means. In fact, for both S infants and mothers, all mean scores were within one standard deviation of data-bank means. Thus, there was no evidence that S infants or their mothers were in any way unusual or abnormal prior to separation manipulations.[3]

Behaviors During Repetitive Separations

Mean levels of prominent behaviors displayed by infants and their mothers during the 4-day separations are presented in Figure 1. Infants displayed much higher levels of vocalization and locomotion—behaviors characteristic of separation protest (e.g., Kaufman & Rosenblum, 1967; Suomi, 1976)—than did mothers. The mothers, by contrast, were largely passive throughout the repeated separations.

A summary of the results of the ANOVAs for infant behaviors during the repetitive separation is displayed in the upper half of Table 2. The ANOVA for infant vocalizations revealed a significant main effect of days, with substantially higher levels of Day 1 of separation than on Days 2–4, and a significant effect of separation series, with much higher levels during the first eight separations than during the

[3] The same types of comparisons were made between C group infants (and their mothers) and these data-bank infants (and their mothers) during Weeks 18–22 and Weeks 23–28 of life (Postsep I). Again, no significant differences between groups were disclosed for any categories, indicating that the C group infants and their mothers were also within the normative range of mother–infant dyads housed in this manner at the Primate Laboratory.

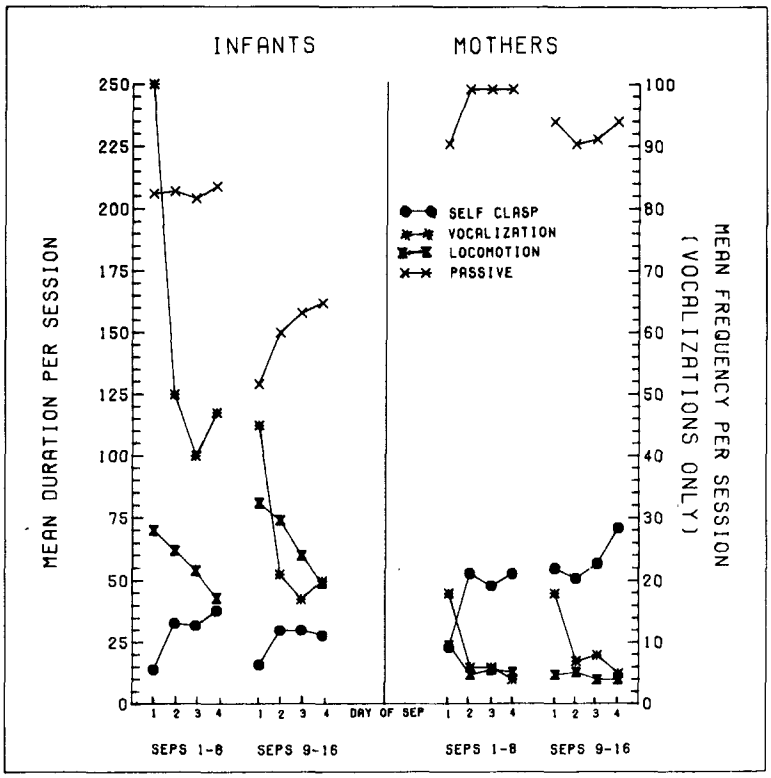


Figure 1. Mean levels of infants' and mothers' scores for vocalization, locomotion, self-clasp, and locomotion categories during separations.

second eight separations. A parallel effect of days was revealed by the locomotion ANOVA, with highest levels on Day 1. The locomotion analysis also yielded a significant Separation Series × Separation Weeks interaction attributable to low overall levels during the first two separations of the first series and to gradually declining levels throughout the second series of eight separations, as well as a significant Weeks × Days interaction, due primarily to greater differences between separation days during early weeks of each separation series than during later weeks of each series. Taken as a whole, these results indicated that infant protest behaviors were highest on the 1st day of each separation and that they tended to decline over repeated separations, especially during the second series of eight separations.

In contrast, the ANOVA for infant self-clasp, a behavior more characteristic of a despair response to separation (e.g., Kaufman & Rosenblum, 1967; Suomi, 1976), yielded only a significant effect of days, with the lowest levels on Day 1. The relatively low levels of this category even on the other separation days (see Figure 1) indicated that despair reactions to separation among the infants were mild at best. Indeed, levels of environmental exploration, hardly a depressive behavior, were highest on the latter days of each separation, especially in the second series, as reflected in a significant days' Effect and Series × Days interaction. The infants became less passive during the second series. Overall, these findings suggest that the infants adapted somewhat to the repeated separations after their initial protests.

The mothers' reactions to the repetitive separations were considerably milder than those of their infants, although there was some evidence of protest (vocalizations higher on Day 1 of each separation than on other days, $p < .01$). Mothers' reactions did not change substantially over the repeated separations. The only significant effect of separation series, weeks, or associated interac-

Table 2
Significant Main Effects and Interactions Disclosed by ANOVAs of Infant Category Scores During Separations and Reunions

Category	Effect	*F*	*df*	*p*	Results of Duncan tests
		Separations			
Vocalization	Series	19.02	1, 3	.025	I > II, $p < .05$
	Days	9.97	3, 9	.005	A > B, C, D, $p < .01$
Locomotion	Days	6.36	3, 9	.025	A > D, $p < .01$ B > D, $p < .05$
	S × W	5.74	7, 21	.001	II 1,2 > I 1,2, $p < .01$ II 1,2 > I 7,8; II 5,6,7,8, $p < .05$
	W × D	2.29	21, 63	.01	1A, 2A > 1D, 2D, 7C, 8D, $p < .05$ 1B, 3B, 4A, > 1D, $p < .05$
Self-clasp	Days	4.80	3, 9	.05	B, C, D > A, $p < .05$
Environmental exploration	Days	8.12	3, 9	.01	B, C, D > A, $p < .05$
	S × D	4.79	3, 9	.05	IIB, C, D > IA, B, $p < .01$ IIB, C, D > IC, IIA, $p < .05$
Passive	Series	59.52	1, 3	.005	I > II, $p < .01$
Stereo	Series	15.05	1, 3	.05	II > I, $p < .05$
	Weeks	2.98	7, 21	.05	8 > 1,2, $p < .05$
		Reunions			
Mother–infant ventral	Observation session	4.77	2, 6	.058	A > C, $p < .05$
Locomotion	Observation session	4.99	2, 6	.053	C > A, $p < .05$
Self-clasp	Observation session	7.34	2, 6	.025	B, C > A, $p < .05$
	S × OS	6.37	2, 6	.05	IIB > IA, B; IIA, C, $p < .01$ IIB > IC, $p < .05$ IC > IA, B, $p < .05$
Vocalization	W × OS	2.15	14, 42	.05	5C > rest, $p < .01$ 2B > 1A, B; 2A; 3A–C; 4A–C; 5A; 6A; 7A–C; 8A, B, $p < .05$

Note. S = series; D = days; W = weeks; OS = observation session. Legend for Duncan tests: Series = I, II; Weeks = 1–8; Observation Sessions = A–D (for separation ANOVAs), A–C (for reunions ANOVAs).

tions disclosed by the ANOVAs was a Series × Days interaction for the passive category, $F(3, 9) = 4.30$, $p < .05$.

Behaviors During Reunions

Figure 2 illustrates behavioral levels displayed by infants and mothers during reunion periods, and significant results from ANOVAs of the infants' reunion behaviors are listed in the lower part of Table 2. The predominant behavior displayed by both infants and mothers during reunions was mother–infant ventral contact. As can be seen in Figure 2a, infants spent between an average of 72% and 100% of their time during reunion sessions in ventral contact with their mothers, with no appreciable decline in this infantile behavior pattern from the first to the second 8 reunions. Indeed, the ANOVA for mother–infant ventral contact yielded only a borderline effect ($p < .058$) for observation sessions, with higher scores during the first than during the last session across all reunions. A corresponding borderline increase over sessions ($p < .053$) was disclosed for infant locomotion, also illustrated in Figure 2. In addition, self-clasping increased over sessions, with significantly higher levels during the second and third observation session than during the first. The self-clasp ANOVA also yielded a significant Reunion Series × Session interaction, largely resulting from higher results in Session 3 than Sessions 1 and 2 during the first series and higher levels in Session 2 than Sessions 1 and 3 during the second series of eight reunions. With the exception of a Weeks × Sessions interaction for vocalization scores (attributable to especially high levels during the third session of Week 5 and the second session of Week 2, for reasons not presently obvious), no other signif-

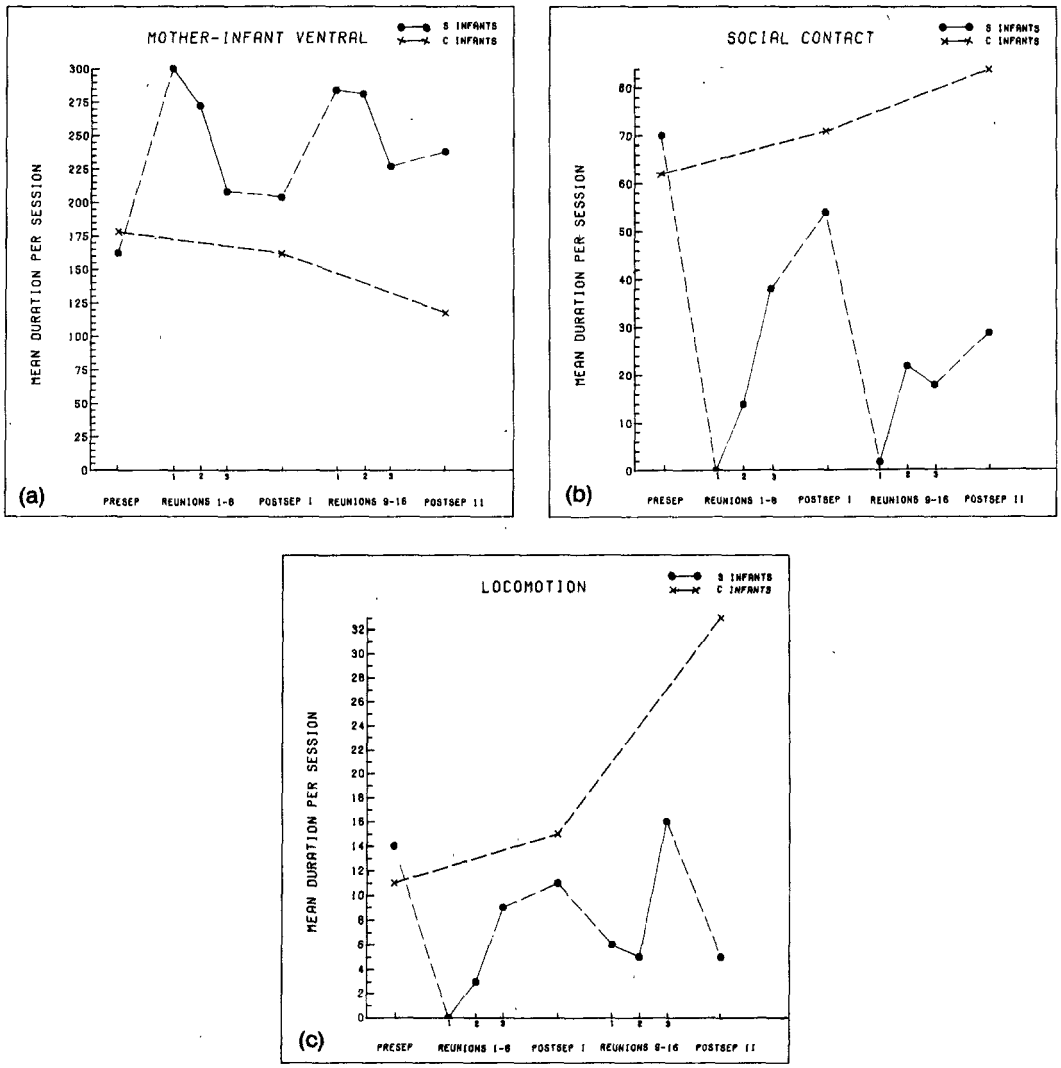


Figure 2. Mean levels of S infants' scores during preseparation and reunion periods and of both S and C infants' scores during postseparation periods for (a) mother–infant ventral contact, (b) social contact, and (c) locomotion. (S = separation; C = control; presep = preseparation; postsep = postseparation. Presep levels plotted for C infants were estimated from laboratory normative data base.)

icant main effects or interactions were disclosed by the ANOVAs for infant reunion scores.

The mother monkeys, like their offspring, spent the vast majority of their time during reunion in mother–infant ventral contact; no significant drops in levels of this behavior over series, weeks, or even sessions was disclosed by the ANOVA. However, a slight increase in social contact with offspring—a less intense and more developmentally advanced form of attachment behavior (Dienske & Metz, 1977; Hansen, 1966; Hinde & Spencer-Booth, 1967)—over reunion series was found, $F(1, 3) = 10.74$, $p < .05$. In addition, a significant weeks' effect was found for passive scores, $F(7, 21) = 2.62$, $p < .05$, attributable to inexplicably high levels during Week 2 of both series of reunions. Significant main effects of sessions were disclosed for vocalization, $F(2, 6) = 11.10$, $p < .01$, and self-clasp, $F(2, 6) = 5.43$, $p < .05$, categories, with higher

levels of vocalization occurring during the first session and higher levels of self-clasp during the second session across reunions. (It should be pointed out that mothers' vocalizations were generally "soliciting" rather than distress vocalizations during these reunion periods.) No other significant main effects or interactions were found in the ANOVAs of the mothers' reunion data.

Postseparation Behaviors

Comparisons between levels of behavior displayed by S monkeys during the two postseparation periods and those displayed by nonseparated C monkeys during comparable chronological periods (Weeks 23–28 and Weeks 37–42 of life) were made via repeated measures ANOVAs, summarized in the upper half of Table 3. Mean levels of major categories during postseparation periods are illustrated in Figure 2.

Significant differences between S and C infants were disclosed for four categories, and in all but one case, the magnitude of these differences increased from Postsep I to Postsep II. As seen in Figure 2, S infants showed a significant increase in levels of infantile mother–infant ventral contact from Postsep I to Postsep II, whereas C infants showed a significant decrease during the same period. The reverse was true for locomotion and social contact. Somewhat surprisingly, C infants had significantly higher levels of self-clasping during both Postsep periods than S infants, although the overall mean levels for

Table 3
Significant Main Effects and Interactions Disclosed by ANOVAs of Infant Category Scores During Postseparation Periods and following Peer Group Formation

Category	Effect	F	df	p	Results of Duncan tests
		Postseparation periods			
Mother–infant ventral	Groups	20.97	1, 6	.005	S > C, $p < .01$
	G × P	9.65	1, 6	.05	SI, II > CII, $p < .01$ SII > CI, $p < .05$ CI > CII, $p < .05$
Social contact	Groups	7.41	1, 6	.05	C > S, $p < .05$
	G × P	8.92	1, 6	.025	CI, II > SII, $p < .01$ CI, II > SI, $p < .05$ SI > SII, $p < .05$
Locomotion	Groups	12.82	1, 6	.05	C > S, $p < .05$
	Periods	6.30	1, 6	.05	II > I, $p < .05$
	G × P	24.24	1, 6	.005	CII > SI, II; CI, p < .01 CI > SII, $p < .05$
Self-clasp	Groups	17.16	1, 6	.01	C > S, $p < .01$
		Peer-group formation			
Social contact	G × P	5.54	3, 18	.01	C2 > S2, 3; C3, $p < .01$
Play & sex	Group	7.88	1, 6	.05	S > C, $p < .05$
Vocalization	Group	8.16	1, 6	.05	C > S, $p < .05$
Locomotion	Period	5.56	3, 18	.01	3 > 2, $p < .01$ 3 > 4, $p < .05$ 1 > 2, $p < .05$
	G × P	9.63	3, 18	.0005	C2 < rest, $p < .01$ S1, 2, 3 > S4, $p < .01$ S3 > C1, 3, $p < .05$
Self-clasp	G × P	6.28	3, 18	.005	C2, S4 > S1, 2, 3; C4, $p < .01$ C2 > C3, $p < .05$
Self-mouth	Period	3.65	3, 18	.05	4 > 1, 3, $p < .05$
Environmental exploration	Group	83.61	1, 6	.0005	C > S
	Period	4.80	3, 18	.01	3 > 1, $p < .01$ 3 > 2, 4, $p < .05$

Note. Legend for Duncan tests: Groups = S (separated infants), C (control infants); Periods = postseparation I, II; Weeks = 1–6; Periods = 1–4.

both groups (17.4 sec vs. 4.8 sec, respectively, per 5-min session) were so low as to be of questionable actual significance. Overall, these results demonstrated that the repetitive material separations had a cumulative effect of retarding the normal development of independence from mother by S infants.

Despite the differences observed between infant groups, there were no significant main effects or interactions revealed by the ANOVAs of mothers' postseparation scores. Borderline Group × Postsep Period interactions were disclosed for the categories of mother–infant ventral contact and exploration; subsequent Duncan tests showed that S mothers had high levels of the former and lower levels of the latter categories than C mothers ($p < .05$) for Postsep II, but not for Postsep I. Thus, group differences between mothers during postseparation periods roughly paralleled those of these infants but were considerably less pronounced.

Final Separation and Peer-Group Housing

In contrast to group differences seen during Postsep I and II, S and C infants displayed surprisingly similar reactions to final maternal separation and initial peer-group formation. Analysis of variance of daily category means for S and C infants during the first 4 days of peer-group housing (Week 43) revealed a significant group difference for exploration (C > S, $p < .05$) and a borderline difference in vocalizations (C > S, $p < .10$), along with significant Group × Days interactions for locomotion (Day 1 S > Day 1 C, Day 3 C > Day 1 S, $ps < .05$) and for exploration (Day 4 C > Day 4 S, $p < .01$). Thus, although C infants displayed slightly greater protest to final (and for them initial) separation from mother than did S infants, by the end of the 1st week of peer housing they were locomoting and exploring more than S infants. However, all these group differences were far from overwhelming in magnitude.

Longer term adjustment to peer-group living was assessed by calculating category means for each S and C subject over Weeks 1–8 (I), 9–16 (II), 17–23 (III), and 24–30 (IV), respectively, of peer-group housing. These means were then subjected to two-way repeated measures ANOVAs, with subject group (S vs. C) as an independent variable and time period (I–IV) as a repeated measure. The results of these ANOVAs are summarized in the lower part of Table 3. Over the 30-week period, C group members explored and vocalized more but played slightly less than did S group members. Both groups showed marginal declines in ventral clinging and significant increases in self-mouthing across time periods. Exploration and locomotion were higher in the third time period than in other time periods. Group × Time Period interactions were found for self-clasp, locomotion, and social contact categories, although the pattern of these interactions defies easy explanation. Taken as a whole, these results suggest that S and C infants displayed roughly comparable reactions to permanent maternal separation and overall adjustment to peer-group living.

Social Preference Tests

The two sets of social preference tests carried out in the Sackett Self-Selection Circus apparatus, when the subjects were approximately 1½ years old, revealed surprising differences between S and C monkeys in relative preferences for their mothers. The results of these tests are presented in Figure 3.

The first preference test, in which each infant chose among its own mother and three strange adult female stimulus monkeys, revealed a significant Groups × Stimulus Animal interaction, $F(4, 24) = 3.01$, $p < .05$. Control infants displayed a strong preference for their own mothers, but all S infants appeared to avoid their own mothers, clearly preferring other adult females. In fact, S infants spent significantly more time in the *empty* choice compartment (no stimulus present) than they did in their mother's compartment (all $ps < .05$).

Results of the second preference test, in which mothers were pitted against fellow peer-group members as stimuli, showed similar group differences. Both S and C infants showed significantly stronger preferences for certain peers than for their mothers, $F(4, 24) = 9.29$, $p < .0005$. However, there was also a significant Groups × Stimulus Animal interaction, $F(4, 24) = 4.39$, $p < .01$, due largely to C monkeys spending significantly more time in proximity to their mothers than S subjects. Indeed, mothers were the *least*

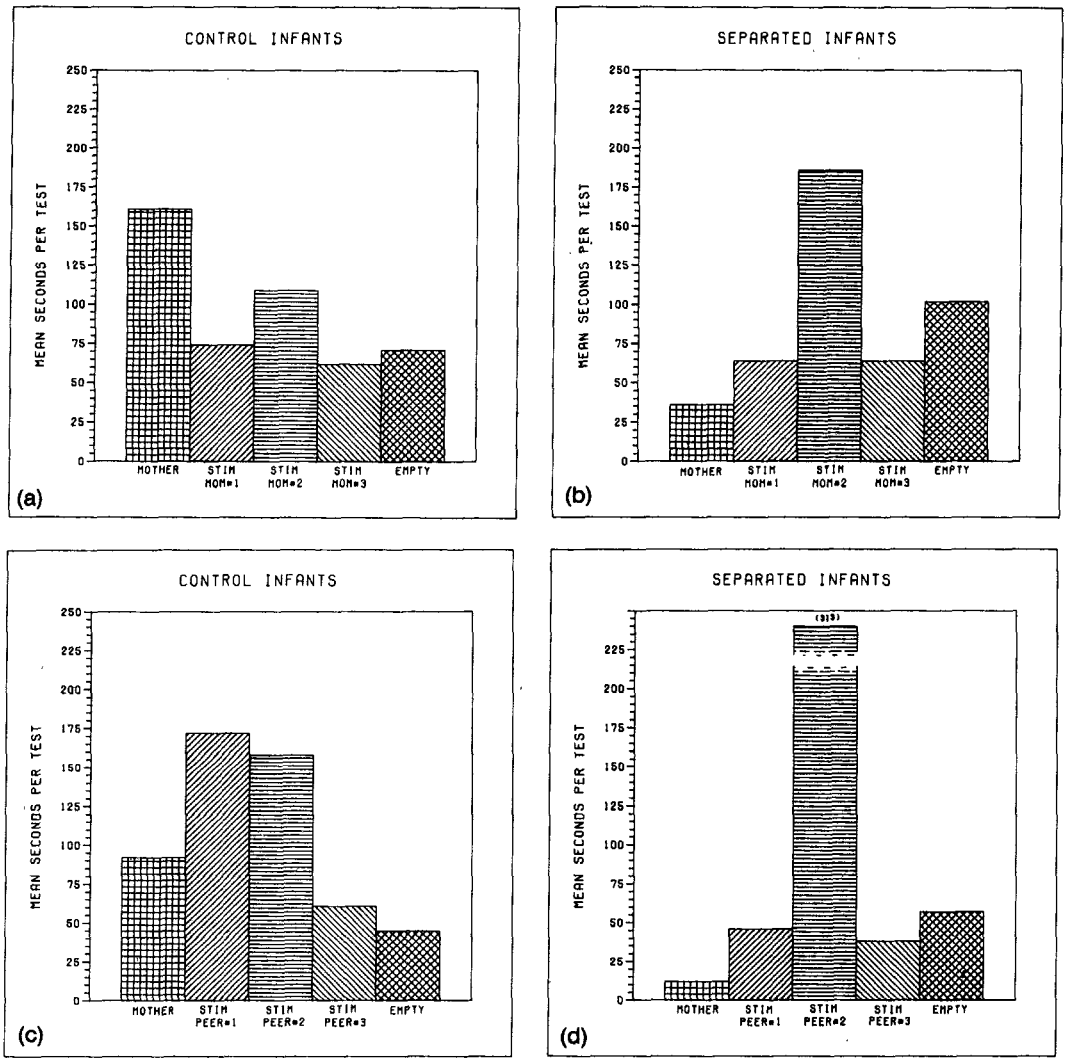


Figure 3. Mean durations of time spent in choice compartment of each stimulus for (a) C infants and (b) S infants during mother versus adult female preference tests, and (c) C infants and (d) S infants during mother versus peer preference tests. (C = control; S = separation.)

preferred of all stimuli for each of the S infants (each S infant again spent significantly more time in the empty compartment than with his or her mother ($p < .05$), adding to the impression that S infants were actually avoiding their own mothers during circus testing.

Discussion

The results of the present experiment are noteworthy in at least four respects. First, they revealed that infants subjected to repetitive 4-day maternal separations displayed species-typical biphasic protest–despair reactions to each of the 16 separations, although the intensity of the protest response declined significantly and the despair was mild at best (none of the infants developed severe depression in response to any of the 16 separations). Second, the repetitive separations appeared to retard the normal development of independence from mother in the separated infants, relative to control infants. Third, in spite of the large differences between separated and control infants in de-

velopment of independence from their mothers, there were few detectable differences in their interactions with peers following a final separation from their mothers. Finally, when given social preference tests more than 7 months following their separation from mother, previously separated infants showed clear-cut avoidance of their mothers, although control infants showed a preference for their own mothers. Each of these four aspects of the results will be discussed in turn.

First, previous studies of single mother–infant separation in rhesus monkey subjects have likewise found clear-cut protest but relatively mild or nonexistent despair when infants were removed from their home cages and housed individually during separations, with a separation of less than 2 weeks' duration (e.g., Hinde & Davies, 1972; Hinde & McGinnis, 1977; Lewis, Young, McKinney, & Kraemer, 1976; Mineka & Suomi, 1978). In addition, other studies examining mothers' reactions to single infant separations have found, as we did, that the mothers' reactions are considerably less intense than those of their offspring (e.g., Hinde & McGinnis, 1977; Jensen, 1968; Jensen & Tolman, 1962). Thus these two aspects of the present results are consistent with reports from previous studies of both infant and mother rhesus reactions to single, short-term separations. On the other hand, our results differ from those of previous studies involving single or widely spaced mother–infant separations in that we found significant decreases in the infants' protest response over many repeated, closely spaced separations. In this regard our results most closely resemble those of Suomi et al. (1970), who also found decreases in protest over repeated peer separations in infants, but no changes in despair. The failure to find an increase in despair in either of these studies contrasts with the results of Mineka et al. (1981) for peer-reared juvenile and adolescent monkeys and, therefore, suggests the possibility that the dramatic increases in despair seen in that study were more a function of the age of the subjects rather than the type of attachment object involved in the separations.

From a theoretical standpoint, the apparent decline in protest over repeated separations (and corresponding lack of despair) seems generally most consistent with the predictions of a social-learning perspective that predicts habituation of the response to separation. However, because this approach does not make explicit differing predictions regarding changes in protest versus despair, it is not clear that it really has a predictive advantage over learned helplessness or opponent-process theories that predict both a decline in protest (supported) and an increase in despair (not supported). Finally, since an attachment-theory perspective does not seem to make an explicit prediction regarding changes in the response to repeated separations, it is neither supported nor refuted by this aspect of our results.

The second noteworthy aspect of our results was our finding that repetitive separations seemed to retard the growth of normal independence from mother. As was shown in Figure 2, levels of mother–infant ventral contact for S infants during reunions were far greater than their levels of ventral contact during preseparation and postseparation periods; these levels were also much higher than levels of nonseparated C infants during comparable chronological periods. Correspondingly, S infants' levels of locomotion and more developmentally advanced social contact increased during the same time periods over their own preseparation baseline and chronologically equivalent control levels. These same tendencies carried over into the postseparation periods: Relative to nonseparated control monkeys, the separated subjects appeared to be developmentally retarded, paralleling the infant peer-reared and repetitively separated monkeys in the Suomi et al. (1970) study.[4]

[4] One note of caution in interpreting these group differences in reunion and postseparation behavior as being the result of separations per se is in order. S and C infants differed not only in their experience with separations but also in their experience with the stressful separation procedure itself and with the stress of living in novel environments during the separations. However, we believe that these factors may have played at most a minor role in the observed group differences, based on the following evidence. First, Levine and his colleagues (e.g., Coe & Levine, 1981) have repeatedly observed that the stressful effects of the separation itself are minimal if the infant can buffer its response by immediate reaccess to the mother. And second, Hinde and McGinnis (1977),

These findings of increased infant dependence on mother can be interpreted as being consistent with Bowlby's (1969, 1973) predictions regarding the emergence of anxious attachment, a syndrome occurring when an infant has no confidence that his or her attachment object will remain available and accessible. As a result, the infant tends to maintain close proximity to the mother in order to ensure continued access. Unfortunately, our scoring system did not allow us to assess whether it was primarily the mothers or the infants who were most responsible for the high level of mutual ventral contact (cf. Hinde & White, 1974). Since group differences in mothers' behaviors were considerably less pronounced than group differences in infants' behavior, it seems likely that it was more the infants' responsibility than the mothers', but without a different scoring system we cannot be sure. Thus, although our results are consistent with the notion that the separated infants became increasingly anxiously attached over time, we cannot be certain that this is an accurate interpretation. Certainly, however, these findings are not consistent with social-learning or opponent-process approaches, which would predict increased infant independence from the mother as a result of separations.

A third interesting aspect of our results was the paucity of group differences during the last 7 months of the experiment, when the monkeys were housed together in peer groups. Although striking differences existed between S and C infants during Postsep II when they were housed with their mothers, there were surprisingly few group differences in reaction to final maternal separation and subsequent peer-group housing. Control infants displayed somewhat greater protest immediately following this separation than did S infants, as might be expected, but this difference was transient. The few group differences and Group × Time Period interactions that did emerge during the 30-week period of group housing were small in magnitude and could probably be accounted for as much by the different sex composition of the two groups as by the different separation histories.[5] These results suggest that carry-over effects from previous separations did not greatly influence S monkeys' reactions to the final separation, peer-group formation, or long-term peer-interaction patterns. In other words, in the absence of their mothers, previous behavioral differences between S and C monkeys tended to disappear. (See also Suomi, Collins, Harlow, & Ruppenthal, 1976, for similar findings.) It is, of course, somewhat risky to draw strong conclusions on the basis of a failure to find group differences, because the possibility always exists that the measurement techniques were insensitive or that "real" differences might have been "masked" in the particular situations in which the subjects were being observed.

Finally, perhaps the most surprising finding of this study was that infants subjected to repetitive maternal separations clearly appeared to avoid their mothers when re-exposed to them in the circus after a 7-month absence. Control infants, in contrast, preferred their mothers to strange females and did not avoid them in the presence of agemates, replicating preference patterns exhibited by normally reared juveniles in previous studies (cf. Sackett, 1970). These findings are especially striking in light of the heightened levels of attachment behavior displayed during reunion and postseparation periods by S infants relative to C infants and the apparent lack of major group differences once both S and C infants were living with peers away from their mothers. Moreover, previous studies of preference patterns of young rhesus monkeys whose relationships with their mothers were not normal, either because their mothers were abusive or because they

among others, reported minimal stressful aftereffects of a procedure in which mothers and infants were separated from their social group and placed together in a novel environment.

[5] Although sex differences in rhesus monkey mother–infant interactions tend to be minor (e.g., Mitchell, 1968; White & Hinde, 1975), sex differences in peer interactions during the first 1½ years of life are considerably more substantial (e.g., Goy, Wallen, & Goldfoot, 1974; Harlow, & Lauersdorf, 1974; Suomi, 1979). Thus, in the present study, the slightly higher play levels shown by S subjects would be predicted simply from the fact that all four S infants were male, whereas two males and two females were in the C group.

had experienced repeated *brief* (2 hour) separations during infancy, have reported preferences for mother to be exaggerated rather than reduced or reversed (Sackett, 1970). Because the proximate basis for the apparent aversion by S infants toward their mothers cannot be determined from the circus test data alone,[6] explanations for its occurrence are presently limited to speculation. Nevertheless, this phenomenon is reminiscent of the avoidance behavior displayed during reunions in the Ainsworth Strange Situation (Ainsworth, Blehar, Waters, & Wall, 1978) by Type A avoidant human babies who are often described as insecurely attached—a description that is probably also accurate for our monkeys (Main, 1977, 1980). Main has developed an elaborate explanation of the human infant's avoidance behavior as representing an adaptive coping response, and a parallel explanation of our monkeys' behavior does not seem unreasonable (cf. Kaufman, 1977; and Rosenblum & Plimpton, 1980, for similar accounts of coping behavior in separated macaque infants who have been visually but not physically reunited with their mothers).

What relevance, if any, do these findings from studies of repetitive maternal separations in rhesus monkeys have for considerations of repetitive short-term separations among human mothers (and fathers) and infants? We clearly acknowledge that cross-species generalizations should be made with great caution and according to well-thought-out principles and that true one-to-one matches of phenomena between species are relatively rare (Suomi & Immelmann, 1983). Furthermore, the living conditions of our monkeys during separations are not analogous to the living conditions of most human infants during parental separations, and it seems unlikely that exactly the same pattern of results would occur if our infant monkeys had been with peers or other adults during their separations, as human infants usually are. Nevertheless, the present monkey data raise some issues that might profitably be kept in mind when pondering possible consequences of frequent parental separations in young children. In particular, even though infants may appear to "get used" to repeated separation from attachment objects, their patterns of behavior when they are with these attachment objects may well be affected. Our separated monkey infants decreased their protest behavior and never became seriously depressed during successive periods of separations; however, the differences in their behavior from that of control infants when with their mothers became increasingly dramatic as the study progressed. Moreover, although group differences seemed to disappear in the absence of the mothers during subsequent peer-group living, these differences reappeared in force during preference tests involving the mothers. Thus, some effects of repetitive maternal separations may be "masked" in certain settings, even though they are readily apparent in other situations.

[6] Since the mother's behavior cannot be observed during circus testing and since the mother and infant can see each other through the plexiglas, it is of course possible that the mother's behavior might have influenced the infant's social choices. However, the same potential problem has existed in other studies in which mothers who would be expected to engage in rejecting behavior (e.g., abusive mothers) nonetheless had infants who *preferred* them over other social objects. Thus, this explanation of our results does not seem too likely.

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